

第21讲. 全微分方程. 曲面积分.

作业: P166. 16①. ④. 17①. ③.

P174. 1. 2. 3. 4.

复习: 1. 若存在 $u(x, y)$ s.t. $du = p dx + q dy$, 则称 $p(x, y) dx + q(x, y) dy = 0$ 为全微分方程2. 若 $p dx + q dy = 0$ 不是全微分方程. 但存在非零函数 $\mu(x, y)$ s.t. $\mu p dx + \mu q dy = 0$ 为全微分方程. 则称 $\mu(x, y)$ 为 $p(x, y) dx + q(x, y) dy = 0$ 的积分因子.

单连通区域

$$\frac{\partial(\mu p)}{\partial y} = \frac{\partial(\mu q)}{\partial x}$$

$$\mu_y' p + \mu p_y' = \mu_x' q + \mu q_x'$$

① $\mu_y' = 0$.

$$\mu p_y' = \mu_x' q + \mu q_x'$$

$$\mu(p_y' - q_x') = \mu_x' q$$

$$\frac{\mu_x'}{\mu} = \frac{p_y' - q_x'}{q} \quad \text{仅仅是 } x \text{ 的函数}$$

$$\ln|\mu(x)| = \int \frac{p_y' - q_x'}{q} dx$$

$$\mu(x) = e^{\int \frac{p_y' - q_x'}{q} dx} \quad \text{为 } p dx + q dy = 0 \text{ 的积分因子}$$

② $\mu_x' = 0$.

$$\mu_y' p + \mu p_y' = \mu q_x'$$

$$\mu(p_y' - q_x') = -\mu_y' p$$

$$\frac{\mu_y'}{\mu} = \frac{p_y' - q_x'}{-p} \quad \text{仅仅是关于 } y \text{ 的函数}$$

$$\mu(y) = e^{\int \frac{p_y' - q_x'}{-p} dy} \quad \text{为 } p dx + q dy = 0 \text{ 的积分因子}$$

例1. 求 $(y \cos x - x \sin x) dx + (y \sin x + x \cos x) dy = 0$ 通解.解: 令 $p(x, y) = y \cos x - x \sin x$, $q(x, y) = y \sin x + x \cos x$

$$\frac{\partial p}{\partial y} = \cos x \neq \frac{\partial q}{\partial x} = y \cos x + \cos x - x \sin x$$

$$\frac{\partial r}{\partial y} = \cos x \neq \frac{\partial u}{\partial x} = y \cos x + \cos x - x \sin x$$

$$p_y' - q_x' = x \sin x - y \cos x = -p$$

$$\therefore \frac{p_y' - q_x'}{-p} = 1$$

$$\therefore u(y) = e^y \text{ 为 } p dx + q dy = 0 \text{ 的积分因子.}$$

$$\therefore e^y p dx + e^y q dy = 0 \text{ 全微分方程.}$$

$$\text{即 } e^y (y \cos x - x \sin x) dx + e^y (y \sin x + x \cos x) dy = 0 \text{ 全微分方程}$$

$$\text{下面用不定积分法求 } e^y (y \cos x - x \sin x) dx + e^y (y \sin x + x \cos x) dy \text{ 的一个原函数.}$$

$$u(x, y) = \int e^y (y \cos x - x \sin x) dx + \varphi(y)$$

$$= e^y y \sin x - e^y (-x \cos x + \sin x) + \varphi(y)$$

$$\int x \sin x dx = - \int x d \cos x = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$$

$$e^y q = u_y' = e^y y \sin x + e^y \sin x + e^y (x \cos x - \sin x) + \varphi'(y)$$

$$e^y (y \sin x + x \cos x)$$

$$\therefore \varphi'(y) = 0$$

$$\therefore \varphi(y) = C.$$

$$\therefore p dx + q dy = 0 \text{ 的通解为 } e^y y \sin x + e^y (x \cos x - \sin x) = C.$$

$$\text{例2. 求 } (x+y) dx + (y-x) dy = 0 \text{ 的通解.}$$

$$x dx + y dx + y dy - x dy = 0$$

$$(x dx + y dy) + (y dx - x dy) = 0$$

$$\frac{1}{x^2+y^2} (x dx + y dy) = 0$$

$$\frac{1}{2} d \ln(x^2+y^2) = \frac{x}{x^2+y^2} dx + \frac{y}{x^2+y^2} dy = \frac{x dx + y dy}{x^2+y^2}$$

$$\frac{1}{2} d(\ln(x^2+y^2)) = \frac{x}{x^2+y^2} dx + \frac{y}{x^2+y^2} dy = \frac{x dx + y dy}{x^2+y^2}$$

$$\frac{1}{x^2+y^2} (y dx - x dy) = 0.$$

$$d(\arctan \frac{x}{y}) = \frac{y}{x^2+y^2} dx + \frac{-x}{x^2+y^2} dy = \frac{y dx - x dy}{x^2+y^2}$$

$$\frac{d(\arctan \frac{x}{y})}{dx} = \frac{\frac{1}{y}}{1 + \frac{x^2}{y^2}} = \frac{y}{x^2+y^2}$$

$$\frac{d(\arctan \frac{x}{y})}{dy} = \frac{-\frac{x}{y^2}}{1 + \frac{x^2}{y^2}} = \frac{-x}{x^2+y^2}$$

$$\therefore d(\frac{1}{2} \ln(x^2+y^2) + \arctan \frac{x}{y}) = 0.$$

$$\therefore \frac{1}{2} \ln(x^2+y^2) + \arctan \frac{x}{y} = C \text{ 为 } (x+y) dx + (y-x) dy = 0 \text{ 通解.}$$

$$\text{验证, } x=0, y=0 \text{ 也是 } (x+y) dx + (y-x) dy = 0 \text{ 解.}$$

$$\textcircled{1} d(xy) = y dx + x dy$$

$$y dx + x dy = 0.$$

$$\textcircled{2} d(\frac{x}{y}) = \frac{1}{y} dx - \frac{x}{y^2} dy = \frac{y dx - x dy}{y^2}$$

$$\textcircled{3} d(\ln \frac{x}{y}) = \frac{1}{x} dx - \frac{1}{y} dy = \frac{y dx - x dy}{xy}$$

$$\frac{\frac{y}{x} \cdot \frac{1}{y}}{\frac{y}{x} \cdot (-\frac{x}{y^2})}$$

$$\textcircled{4} d(\ln |\frac{x-y}{x+y}|) = \frac{2y}{x^2-y^2} dx + \frac{-2x}{x^2-y^2} dy = \frac{2(y dx - x dy)}{x^2-y^2}$$

$$\frac{\frac{x+y}{x-y} \cdot \frac{2y}{(x+y)^2}}{\frac{x+y}{x-y} \cdot \frac{-2x}{(x+y)^2}} = \frac{2y}{x^2-y^2}$$

$$\frac{\frac{x+y}{x-y} \cdot \frac{-2x}{(x+y)^2}}{\frac{x+y}{x-y} \cdot \frac{-2x}{(x+y)^2}} = \frac{-2x}{x^2-y^2}$$

$$\begin{aligned} \textcircled{5} d(x \ln y - y \ln x) &= (\ln y - \frac{y}{x}) dx + (\frac{x}{y} - \ln x) dy \\ &= \ln y dx + \frac{x}{y} dy - (\frac{y}{x} dx + \ln x dy) \end{aligned}$$

$$= \ln y dx + \frac{x}{y} dy - \left(\frac{y}{x} dx + \ln x dy \right)$$

求 $\ln y dx + \frac{x}{y} dy - \left(\frac{y}{x} dx + \ln x dy \right) = 0$ 通解.

$$\downarrow$$

$$x \ln y - y \ln x = c.$$

第 I 型
= 曲面面积, —— 关于面积, 曲面积分.

1. 物理来源: 求光滑曲面片质量.

$S \subset \mathbb{R}^3$ 光滑曲面片. $\forall (x, y, z) \in S$. $\rho(x, y, z)$ 质量面密度. 求 S 质量.

$$\forall T = \{ S_1, S_2, \dots, S_n \}. \quad \Delta S_i \text{ 表示 } S_i \text{ 面积.}$$

$$\|T\| = \max \{ \text{diam } S_i \mid i=1, 2, \dots, n \}$$

$$\forall (x_i, y_i, z_i) \in S_i.$$

$$M = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n \rho(x_i, y_i, z_i) \Delta S_i$$

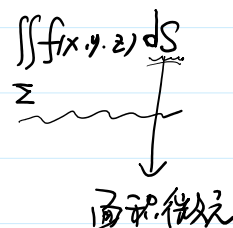
定义 1. 设 $f(x, y, z)$ 定义在光滑曲面片 $\Sigma \subset \mathbb{R}^3$ 上. $\forall T = \{ \bar{S}_1, \bar{S}_2, \dots, \bar{S}_n \}$.

ΔS_i 表示 $\bar{S}_i$ 面积. $\|T\| = \max \{ \text{diam } \bar{S}_i \mid i=1, 2, \dots, n \}$.

$$\forall (x_i, y_i, z_i) \in \bar{S}_i.$$

若 $\lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta S_i$ 存在.

且与划分方式无关, 与点取法无关. 则 $f(x, y, z)$ 在 Σ 上可积, 记 $\iint_{\Sigma} f(x, y, z) dS$



Σ 积分曲面.

定理 1. ① 设曲面 $\Sigma: z = f(x, y) \in C^1(D)$. $D \subset \mathbb{R}^2$ 有界闭区域.

若 $g(x, y, z) \in C(\Sigma)$. 则 $\iint_{\Sigma} g(x, y, z) dS = \iint_D g(x, y, f(x, y)) \sqrt{1 + f_x'^2 + f_y'^2} dx dy$

② 曲面 $\Sigma: X = X(y, z) \in C^1(D)$.

若 $g(x, y, z) \in C(\Sigma)$, 则

$$\iint_{\Sigma} g(x, y, z) dS = \iint_D g(x(u, v), y(u, v), z(u, v)) \sqrt{1 + x_u'^2 + x_v'^2} du dv.$$

定理 2. 设光滑曲面片 Σ : $\begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad (u, v) \in D.$

若 $h(x, y, z) \in C(\Sigma)$, 则

$$\iint_{\Sigma} h(x, y, z) dS = \iint_D h(x(u, v), y(u, v), z(u, v)) \sqrt{EF - G^2} du dv$$

$$E = x_u'^2 + y_u'^2 + z_u'^2$$

$$EF - G^2 = A^2 + B^2 + C^2$$

$$F = x_v'^2 + y_v'^2 + z_v'^2$$

$$A = \det \frac{\partial(y, z)}{\partial(u, v)}, \quad B = \det \frac{\partial(z, x)}{\partial(u, v)}, \quad C = \det \frac{\partial(x, y)}{\partial(u, v)}$$

$$G = x_u' x_v' + y_u' y_v' + z_u' z_v'$$

例 1. 设 Σ, Σ' 为光滑曲面片, 正交变换 $U: \Sigma \rightarrow \Sigma'$.

$$\text{则 } \forall f(x, y, z) \in C(\Sigma), \quad \iint_{\Sigma} f(x) dS = \iint_{\Sigma'} f(U^T Y) dS$$

$$X = (x, y, z)^T.$$

$$UX = Y$$

$$\Sigma : \begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad (u, v) \in D.$$

$$X = \begin{pmatrix} x(u, v) \\ y(u, v) \\ z(u, v) \end{pmatrix}$$

$$U: \Sigma \rightarrow \Sigma'$$

$$\Sigma' : Y = UX$$

$$X_u' = \begin{pmatrix} x_u'(u, v) \\ y_u'(u, v) \\ z_u'(u, v) \end{pmatrix}$$

$$dS = \sqrt{EF - G^2} du dv$$

$$\rightarrow Y_u' = UX_u'$$

$$E = x_u'^2 + y_u'^2 + z_u'^2 = X_u'^T X_u'$$

$$F = (x_v')^T x_v'$$

$$G = X'_u \cdot X'_v = \langle X'_u, X'_v \rangle$$

$$\Sigma' \quad ds' = \sqrt{E'F' - G'^2} du dv = \sqrt{EF - G^2} du dv = ds.$$

$$E' = (Y'_u)^T Y'_u = (X'_u)^T U^T U X'_u = (X'_u)^T X'_u = E$$

$$F' = F$$

$$G' = G$$

例 2. 设 $f(u)$ 连续. 且 $\Sigma: x^2 + y^2 + z^2 = 1$.

$$\text{证明: } \iint_{\Sigma} f(ax+by+cz) ds = 2\pi \int_{-1}^1 f(\lambda u) du. \quad \text{其中 } \lambda = \sqrt{a^2+b^2+c^2}$$

$$\text{证: } u = \frac{1}{\lambda}(ax+by+cz) = \left(\frac{a}{\lambda} \quad \frac{b}{\lambda} \quad \frac{c}{\lambda}\right) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\text{正交矩阵 } U = \begin{pmatrix} \frac{a}{\lambda} & \frac{b}{\lambda} & \frac{c}{\lambda} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix}.$$

$$U^T U = U U^T = I.$$

$$U \begin{pmatrix} \lambda \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

$$u^2 + v^2 + w^2 = \underbrace{(u, v, w)}_{(x, y, z)} \begin{pmatrix} \lambda \\ y \\ z \end{pmatrix} = (x, y, z) U^T U \begin{pmatrix} \lambda \\ y \\ z \end{pmatrix} = x^2 + y^2 + z^2 = 1.$$

$$\iint_{\Sigma} f(ax+by+cz) ds = \iint_{u^2+v^2+w^2=1} f(\lambda u) ds$$